

Evaluating design-stage manufacturability decisions against the resolution of production

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Abstract

Design-stage manufacturability decisions increasingly rest on simulation and machine learning, yet whether a prediction is fit to decide is rarely evaluated against production. A framework is proposed whose unit, the production floor, is the difference that drift and scatter produce between two batches of one design within a run. A rule returning meets, fails or trial is characterised by a decisive distance in floors, beyond which at least 95 % of its verdicts are correct, and a safe distance, beyond which at most 5 % are wrong, for each relation of a new design to the produced evidence: a variant, a new process setting or a new family. A procedure turns them into a guard band for single verdicts. Thirteen rules were evaluated at a 95 % conformance level on open data of 18 deep-drawn design–process alternatives of 500 parts each, mostly process settings on one tool, with matched simulations; one floor of draw-in is 0.05 mm. With a produced sibling the nearest produced setting decided within 3.4 floors; for a new process setting no rule decided closer than 8.5 floors; in two transfers to a new family none was safe closer than 16. Within a family the choice of rule mattered more than the relation, and requirements at a performance index of 1.33 lay inside every decisive distance. With six to nine produced settings per fit, the simulation, as used, helped only across families, and learned models added abstention rather than accuracy. The framework gives evaluations of AI-based manufacturability support a production-referenced standard.

Keywords: design for manufacturing, machine learning, design decision making, verification and validation, operating characteristic, deep drawing

1 Introduction

Whether a design alternative can be made to its requirements is decided long before it is made. In sheet-metal forming, as in most processes, the decision rests on a finite-element simulation (Banabic, 2010): the team compares the predicted characteristics with the requirements and releases the alternative, changes it or sends it to a physical trial. Simulation has moved such decisions forward (Thomke, 1998), and machine learning makes the loop faster: surrogates replace simulations, models learn manufacturability from earlier designs, and checks are automated (Yan et al, 2026; Ghadai et al, 2018; Zhang et al, 2018). Neither has changed the question the team has to answer: *can this prediction be relied on to decide?* A simulation that is right on average can be wrong for the alternative at hand, and a model learned on earlier products may be asked about a design unlike any of them. Whitney argued in this journal that mechanical design cannot be verified by simulation the way integrated circuits are (Whitney, 1996); production variation, with which development organisations still contend (Thornton et al, 2000), adds a further reason. Whether a simulation-based decision is credible is in the end a property of production, and production evidence is what is scarce at the design stage.

This paper treats the design-stage manufacturability decision itself as the object of study. How can the fitness of a design-stage prediction for such a decision be expressed on a common production-referenced scale, for any predictor and characteristic (RQ1)? How does it depend on the amount of production evidence and on how the new design relates to what has been produced (RQ2)? And what do simulation and machine learning contribute once production evidence exists (RQ3)? Decision-based design (Hazelrigg, 1998; Lewis et al, 2006), design margins (Eckert et al, 2019; Brahma and Wynn, 2020) and the planning of verification and testing (Thomke and Bell, 2001; Salado and Kannan, 2018; Tahera et al, 2019) all need to know how far a prediction can be relied on before a test is cheaper than a decision.

The framework proposed here has the *production floor* as its unit: the difference that drift and scatter produce between two batches of consecutive parts of one design within a run, a floor in the sense of a noise floor rather than of the shop floor. A rule that turns the design-stage evidence into a verdict (the alternative *meets* the requirement, *fails* it, or goes to a physical *trial*) is characterised by two operating characteristics in floors, its *decisive distance* and its *safe distance*, for each *evidence relation* between a new design and the produced alternatives, and a procedure turns them into a guard band for single verdicts. The decisions in scope are those of a team that has produced variants of a design family, in process planning and tryout and when production knowledge is carried to a new family; early product design is covered only by the new-family relation.

The evaluation uses open data: 9,000 deep-drawn and cut cups of dual-phase steel, produced as 18 design–process alternatives of 500 consecutive parts each (Baum and Heinzelmann, 2026), with finite-element simulations of the same process (Baum and Heinzelmann, 2025). Within each of two cup geometries the alternatives differ in blank-holder force and lubrication on one tool, so two of the three relations concern process settings, and the geometries give two transfers between families. Each alternative is judged as a new design, and the verdicts of thirteen rules at a 95 % conformance level

are compared with what its 500 parts did. In the terms of the design research methodology (Blessing and Chakrabarti, 2009), this is a prescriptive study with an initial support evaluation: it shows whether the framework discriminates between rules and evidence situations, not whether design teams decide better with it, and its findings on one dataset are hypotheses for confirmation.

The contributions are the framework (Section 3); an evaluation design that separates the evidence relations by grouped cross-validation (Roberts et al, 2017) and isolates what the simulation adds (Section 4); the empirical answers for deep drawing (Section 5); and guidance and a reporting standard for evaluating simulation- and AI-based manufacturability support (Section 6). Section 2 positions the work, Section 7 concludes, and Appendix A gives the notation, the robustness results and computation details.

2 Background and related work

2.1 Manufacturability evaluation and machine learning

Design for manufacturing (DFM) codifies what a process can make into rules, guidelines and cost models (Boothroyd et al, 2010), and quantitative manufacturability models have been used in this journal to drive design decisions (La Trobe-Bateman and Wild, 2003). Machine learning learns manufacturability from CAD models (Ghadai et al, 2018; Zhang et al, 2018; Williams et al, 2019), from process data and simulated design spaces (Yan et al, 2026), and generative models propose designs directly (Regenwetter et al, 2022); transfer between processes and product families remains difficult (Safdar et al, 2024). Learned manufacturability models are typically evaluated by held-out accuracy on labelled designs (Ghadai et al, 2018; Zhang et al, 2018; Williams et al, 2019), not by the decisions they support against the production of the alternatives they are asked about.

2.2 Design decisions, design types and production evidence

Decision-based design casts design as a choice between alternatives with uncertain outcomes (Hazelrigg, 1998; Lewis et al, 2006). How much a team knows about a new alternative depends on the kind of design, variant, adaptive or original (Pahl et al, 2007), organised across products by product-family and platform design (Simpson et al, 2006; Jiao et al, 2007). The evidence relations of this paper are their counterparts for production evidence: a new variant of a produced setting, a new process setting and a new family. Production knowledge enters design through process-capability data (Thornton, 2004; Tata and Thornton, 1999) and tolerance analysis (Chase and Parkinson, 1991). Unresolved uncertainty is absorbed by margins (Eckert et al, 2019; Brahma and Wynn, 2020), and the part of a margin that covers uncertainty is distinct from the part that absorbs change (Eckert and Isaksson, 2017). Tests and prototypes, the alternative to a margin, are modelled as a trade between the costs of tests and of errors (Thomke, 1998; Thomke and Bell, 2001; Salado and Kannan, 2018) and studied as design activities (Tahera et al, 2019; Camburn et al, 2017). When upstream information is fit to act on is the question of overlapped development (Krishnan et al,

1997; Terwiesch et al, 2002), and set-based concurrent engineering codifies production knowledge as feasibility limits for new designs (Sobek et al, 1999). All of these assume a measure of how far a design-stage prediction can be relied on; the framework supplies one in a unit set by production.

2.3 Simulation credibility, calibration and validation

Verification and validation ask whether a model is accurate enough for its intended use, and the distinction between its validation and application domains (Oberkampf and Roy, 2010) is the question that the evidence relation asks of a design; risk-informed credibility assessment scales the required evidence with the consequence of a wrong decision (ASME, 2018). Bayesian calibration adds a discrepancy model and calibration parameters to a simulator (Kennedy and O’Hagan, 2001), multi-fidelity models scale a cheap model to an expensive one (Kennedy and O’Hagan, 2000), and omitting the discrepancy biases the parameters (Brynjarsdóttir and O’Hagan, 2014); where a model may be relied on at all is the question of its valid input domain (Malak and Paredis, 2010). Conformal prediction gives distribution-free intervals with finite-sample coverage under exchangeability (Lei et al, 2018; Barber et al, 2021), also under covariate shift (Tibshirani et al, 2019) and for the control of decision risks (Angelopoulos et al, 2024). In sheet-metal forming, friction (Hol et al, 2012) and the unloading behaviour of the material (Yoshida and Uemori, 2002) are among the main sources of simulation error; simulation datasets of deep drawing support surrogate and transfer learning (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025), and the deviation between simulation and parts has been decomposed statistically (Baum et al, 2026). How close to a requirement such intervals and accuracy statements can be relied on to decide, and how much production evidence that takes, remains open.

2.4 Decisions with abstention and their operating characteristics

In conformity assessment a measured value proves conformance, proves non-conformance or leaves the decision open, with guard bands set by the measurement uncertainty and the risks of false acceptance and rejection as the measures of a decision rule (ISO, 2017; Joint Committee for Guides in Metrology, 2012). In selection procedures and acceptance sampling the operating characteristic gives the probability of a decision against the true quality, with an indifference zone inside which no decision is required (Bechhofer, 1954). A classifier with a reject option trades errors against abstentions (Chow, 1970; Geifman and El-Yaniv, 2017), learning to defer passes difficult cases to an expert (Madras et al, 2018), and appropriate reliance on automation requires that its reliability be visible to its users (Lee and See, 2004), also in design teams working with AI (Zhang et al, 2021). The decisive and safe distances are operating characteristics of this kind, and the guard band of the procedure is the design-stage counterpart of a conformity-assessment guard band.

Table 1 Positioning among methods that support or evaluate design-stage manufacturability decisions

Approach	Evidence	Output	Scored against production	Abstains
DFM rules and cost models (Boothroyd et al, 2010)	codified process knowledge	guideline compliance, cost	no	no
Design margins (Eckert et al, 2019)	experience, analysis	buffer to the requirement	rarely	no
V&V, credibility assessment (Oberkampf and Roy, 2010; ASME, 2018)	physics model, validation tests	accuracy in physical units; required evidence	on test cases	can require tests
Bayesian calibration (Kennedy and O’Hagan, 2001)	simulations, experiments	predictive distribution	not as decisions	implicitly
Conformal prediction (Lei et al, 2018)	exchangeable calibration data	interval with coverage	coverage; per decision possible on the measured item	yes
Conformity assessment, OC curves (ISO, 2017; Bechhofer, 1954)	measurement and its uncertainty	accept, reject, undecided; risks		yes
Selective classification (Chow, 1970; Geifman and El-Yaniv, 2017)	labelled data	class or reject; risk-coverage	on held-out data	yes
ML manufacturability models (Ghadai et al, 2018; Yan et al, 2026)	historical or simulated designs	class or value	on held-out data	rarely
This paper	simulations, produced alternatives	verdict; decisive and safe distance in floors by evidence relation	yes, per verdict	yes

2.5 Production variation and measurement

Statistical quality control compares short- and long-term variation with tolerances through capability and performance indices (Montgomery, 2019), with release criteria such as a performance index of at least 1.33 (Automotive Industry Action Group, 2006) and time-dependent models for drifting processes (ISO, 2026). Precision experiments define the repeatability limit, the difference that two results under repeatability conditions exceed with a probability of 5 % (ISO, 2025), and measurement systems analysis separates repeatability from stability (Automotive Industry Action Group, 2010). The production floor is a repeatability-type limit of batch centres under within-run conditions; what is new is its use as the unit of a design decision rule. Table 1 places the framework among these approaches: it does not replace calibration or conformal prediction but scores them, as decision rules, against production in a common unit.

3 The framework

3.1 The decision problem and its scope

A design alternative a is a combination of product geometry and process settings that is produced as a series of parts. A quality characteristic c varies over the parts of a series. A requirement is an upper limit R on the characteristic, or on its absolute value for characteristics whose target is zero, that a share p of the parts must meet: alternative a is *adequate* for c when the p -quantile of its parts satisfies $q_p(a, c) \leq R$. Lower or two-sided limits follow by a change of sign or by the absolute deviation from the target; a tolerance of $\pm t$ on a wall angle about its design angle, for example, is the limit $R = t$ on the absolute deviation. At the design stage q_p is unknown. A decision rule r maps the available evidence to an interval $[L_r, U_r]$ for q_p and to a verdict

$$V_r = \begin{cases} \text{meets} & U_r \leq R, \\ \text{fails} & L_r > R, \\ \text{trial} & \text{otherwise.} \end{cases} \quad (1)$$

The three verdicts mirror the decision rules of conformity assessment (ISO, 2017), here for the uncertainty of a design-stage prediction. A rule that returns a point ($L_r = U_r$) always decides. The evidence consists of simulations of a and of the production record of other alternatives, the *calibration alternatives*; it is used by the process planner or tryout engineer for a further variant or process setting of an existing tool, at tool tryout or before production part approval, and by the product designer for a new family. The demonstration decides on $p = 0.95$, which for a normal characteristic corresponds to a performance index of about 0.55. Release criteria of series production, a performance index of 1.33 or more (Automotive Industry Action Group, 2006), concern nonconforming fractions in parts per million that a series of 500 parts cannot estimate without a parametric tail; the framework reaches them through requirements anchored in capability (Section 5.8).

3.2 Production floor, effect-to-scatter ratio and minimum resolvable change

Production evidence comes as series of consecutive parts, each cut into batches of B consecutive parts, and each batch is summarised by a robust centre m (here the 10% trimmed mean). The production floor of characteristic c is

$$F_c = Q_\alpha \{ |m_i - m_j| : i < j \text{ batches of the same alternative} \}, \quad (2)$$

the α -quantile of the difference between two batches of one design. Under a stationary normal model with a standard deviation σ_b between batch means and σ_w between parts, $F_c = 1.96\sqrt{2}\sqrt{\sigma_b^2 + \sigma_w^2}/B$ at $\alpha = 0.95$: the repeatability limit of precision experiments (ISO, 2025, 1994), applied to batch centres under within-run conditions. The reference protocol of this paper is $B = 50$, ten batches of a 500-part run, all pairs of batches and $\alpha = 0.95$, with per-geometry floors where geometries differ. The floor

rests on three assumptions: batches of one design differ only by drift and scatter within a run (A1); the runs from which it is estimated are like those of future production (A2); and the measuring system is repeatable and its resolution small against the floor (A3, checked in Section 5.1).

Four properties matter for its use. It has the unit of the characteristic, so distances in floors are on a common scale for all characteristics and predictors, although equal distances do not mean equal nonconforming fractions, because the ratio of the floor to the scatter of the parts differs between characteristics (Table 3). It is a property of a production protocol: under drift it grows with the time between the batches compared, and it shrinks with B ; σ_b and σ_w are reported so that it can be recomputed for another protocol (Table 3). It contains whatever drifts between batches, the process and the measuring system alike, so without a reference artefact scanned through the series it is a production-and-measurement floor. And it excludes the variation between runs, which makes it a lower bound of the variation of series production. For a new family, whose floor is unknown before it is produced, the floor of the produced family is used. Because the floor resolves batch centres while the requirement concerns a quantile, the reproducibility of the quantile is reported alongside.

Two alternatives a and b are resolvable for c when their effect-to-scatter ratio

$$\text{ESR}_c(a, b) = |\bar{m}_a - \bar{m}_b| / F_c \quad (3)$$

is at least one, $\bar{m}$ being the centre of all parts of an alternative; this is a practical threshold, not a test. For a continuous design or process parameter x with a local sensitivity $s_c = \partial \bar{m} / \partial x$, the minimum resolvable change is $F_c / |s_c|$: a change of x by less moves the centre of the characteristic by less than two batches of one design differ. It is local, a statement about centres, and a change below it can still move the tail of the distribution.

3.3 Decision rules

Thirteen rules span what a design team can do with its evidence (Table 2); each is defined for any set of calibration alternatives, and the first seven carry the results in the main text. The nominal simulation (M0) uses the simulation alone. The bias-corrected simulation (M1) adds the median offset between the calibration alternatives' q_p and their nominal simulation; the scaled simulation (M1s) fits $q_p = a + b s_0$ to them by least squares, a linear multi-fidelity correction (Kennedy and O'Hagan, 2000) that rescales a simulated trend of the wrong size. The envelope rule (M2) adds the median offset and the largest absolute residual of the calibration alternatives to the upper end of the simulation envelope over plausible incoming conditions. M5 models the offset between q_p and the envelope as a Gaussian process (Rasmussen and Williams, 2006) with a 90% predictive interval, whose noise variance is bounded below by the variance between the produced series of one geometry and force, so that a new series of a produced setting is not predicted more precisely than produced series reproduce each other; M5n is the same process of q_p without the simulation. The nearest produced setting (NN) is a capability-database look-up that uses neither simulation nor learning. The six further rules include a friction-calibrated simulation (Kennedy and

323 O’Hagan, 2001) (Mc) and part-level gradient boosting (Friedman, 2001) with a jack-
 324 knife+ interval (Barber et al, 2021), with and without the simulation (M3, M3n). The
 325 nominal simulation and the envelope do not depend on the lubrication pattern, so a
 326 lubrication variant of a produced setting gives M0, M1, M1s, M2 and M5 no simulated
 327 information that distinguishes it from its siblings.

328 The margins are labelled by what they guarantee. With n calibration alternatives,
 329 the largest-residual margin covers a new exchangeable residual with probability $n/(n+1)$
 330 for a fixed offset and at least $(n-1)/(n+1)$ for an estimated one, because the new
 331 residual is then covered whenever it falls within the range of the calibration residuals;
 332 it is the split-conformal quantile (Lei et al, 2018) at the highest attainable level, not a
 333 90 % interval. In general, a distribution-free interval cannot certify a miss rate below
 334 $1/(n+1)$, so a miss rate of 5 % needs at least 19 produced alternatives in the relevant
 335 relation and one of 10 % needs nine. Under exchangeability the jackknife+ interval
 336 covers with probability at least 0.8 at the level of 0.9 used here (Barber et al, 2021),
 337 and the Gaussian-process interval is model-based. The level is a decision variable that
 338 trades decisiveness for safety, as the cost of errors would set the threshold of a reject
 339 option (Chow, 1970); 0.9 is used here, and 0.8 and 0.95 are reported in the robustness
 340 analysis.

341

342 3.4 Operating characteristics: decisive and safe distances

343 A rule is scored by judging each produced alternative as a new design at requirements
 344 placed at a known distance from the truth,
 345

$$346 R = q_p(a, c) + d F_c, \quad (4)$$

347

348 so that a is adequate exactly when $d \geq 0$. For a case (a, c) with interval $[L, U]$, let
 349 $u = (U - q_p)/F_c$ and $v = (q_p - L)/F_c$ be the distances by which the interval excludes
 350 the truth from above and below. At $d = +\delta$ the verdict is correct when $u \leq \delta$ and a
 351 false reject when $-v > \delta$; at $d = -\delta$ it is correct when $v < \delta$ and a false accept when
 352 $-u \geq \delta$. Let $\kappa(\delta)$ and $\omega(\delta)$ be the shares of correct and of wrong verdicts among all
 353 feasible verdicts at $|d| = \delta$: every case gives one verdict on an adequate alternative
 354 and, where $q_p - \delta F_c \geq 0$, one on a failing alternative, and every verdict has equal
 355 weight. As δ grows, failing-side requirements become infeasible for characteristics close
 356 to zero, so the failing side carries half the weight up to a few floors and a third at
 357 about 100 floors (98 of 126 failing-side requirements remain feasible at 16 floors and
 358 63 at 108). The *decisive distance* δ_d of a rule is the smallest δ beyond which $\kappa \geq 0.95$,
 359 and its *safe distance* δ_s the smallest δ beyond which $\omega \leq 0.05$.

360 The distances are exact: κ and ω change only where δ crosses one of the case’s
 361 u , v , $-u$ or $-v$, so the distances are order statistics of these exclusion distances;
 362 for a rule that always decides, $\delta_d = \delta_s$ is close to the 90 % quantile of its absolute
 363 error in floors. Always $\delta_s \leq \delta_d$, and between the two the rule sends designs to trial
 364 rather than misleading; one-sided distances for failing alternatives (false accepts) and
 365 adequate ones (false rejects) matter when the errors cost differently. The distances
 366 are points on operating characteristics over a population of design situations: δ_d is
 367 the analogue of the indifference zone of a selection procedure (Bechhofer, 1954) at a
 368

Table 2 The decision rules; the first seven carry the results in the main text

Rule	Prediction of q_p	Interval	Production evidence
M0 nominal simulation	simulation at nominal sheet thickness and friction	point	none
M1 bias-corrected simulation	M0 plus the median offset	point	q_p
M1s scaled simulation	$a + b$ M0, least squares on the calibration alternatives	point	q_p
M2 envelope + residual margin	upper end of the envelope plus the median offset	largest residual	q_p
M5 GP correction of the envelope	envelope plus a Gaussian-process offset over the descriptors	GP, 90 %	q_p
M5n GP of q_p	Gaussian process of q_p over the descriptors	GP, 90 %	q_p
NN nearest produced setting	mean q_p at the nearest produced process setting of the family	point	q_p
M0w process-window worst case	lowest to highest simulated value over the incoming conditions	range	none
Mc friction-calibrated simulation	simulation with friction calibrated on the draw-in, plus the median offset	point	centres, q_p
M3 simulation + boosting	matched simulation plus a boosting model of the discrepancy, part level	jackknife+	parts
M3n boosting, no simulation	boosting model of the characteristic, part level	jackknife+	parts
M1n median of the produced	median q_p of the calibration alternatives	point	q_p
M2n median + residual margin	M1n with the largest-residual margin	largest residual	q_p

Descriptors: geometry indicator, blank-holder force and lubrication rank (the order of the oil amount). GP: noise variance bounded below by the variance between the series of one geometry and force. M3n is named M4 in the released code; M6, a conformalised GP, appears only in the robustness analysis.

probability of correct selection of 0.95, and the pair plays the role of the acceptable and rejectable quality levels of an acceptance plan. A trial is itself not error-free: a short run reproduces the 95th percentile only to about one floor (Section 5.1), so no decision can be relied on closer to the requirement than the trial that would replace it.

3.5 Evidence relation, calibration budget and applicability guard

How a new design relates to the produced alternatives is part of the decision situation. With a *sibling*, an alternative at the same process setting has been produced with another variation that production may not resolve; a *new process setting* lies between produced settings (interpolation) or outside them (extrapolation); a *new family* shares no produced alternative. Beyond a factorial, the relation can be graded by a distance in descriptor space or by the effect-to-scatter ratio between the predicted characteristic and the nearest produced alternative (Section 6.5). The calibration budget is the dependence of the distances on the number k of produced alternatives, over all subsets

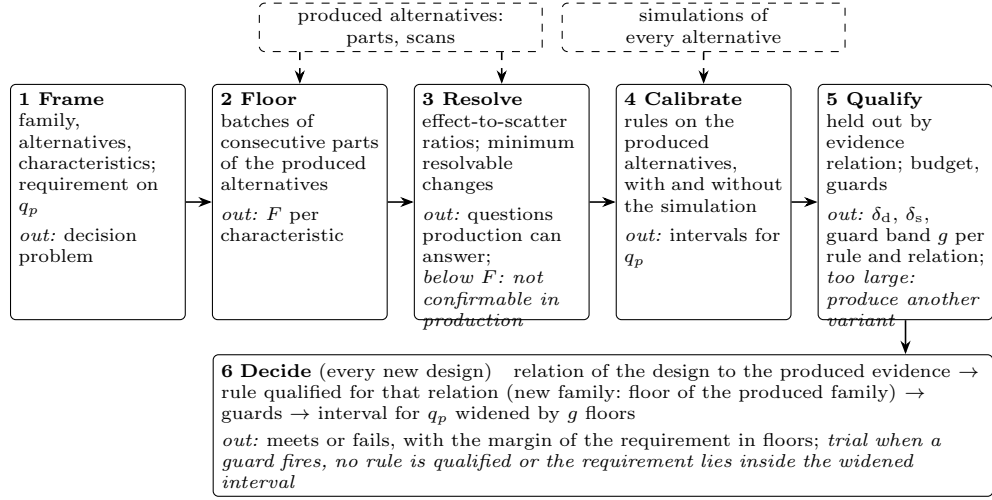


Fig. 1 The procedure: steps 1–5 once per design family, step 6 for every new design; dashed boxes are the evidence it uses

of size k classified by relation. An applicability guard returns *trial* when the new design lies outside the calibration: its geometry is not calibrated (family guard), a descriptor lies outside the calibrated range (range guard), or its simulated q_p lies outside the calibration alternatives' simulated values (novelty guard), in the spirit of a valid input domain (Malak and Paredis, 2010).

3.6 Procedure

Figure 1 turns the framework into six steps: steps 1 to 5 once per design family, repeated when production evidence is added, and step 6 for every new design. Step 1 frames the decision (family, alternatives, characteristics, form of the requirement and conformance level); step 2 measures the floor with a stated protocol; step 3 states which differences between alternatives, and which local changes of a parameter, production can resolve; and step 4 calibrates the candidate rules with and without the simulation. Step 5 qualifies them by their distances for each evidence relation and fixes for each rule and relation a *guard band* g : the smallest observable margin between the predicted interval and the requirement beyond which at least 95 % of the decided verdicts were correct, for requirements within 20 floors of the truth. If the distances are too large for the requirements of coming designs, another variant is produced. Step 6 establishes the relation of the new design, applies the rule qualified for it and the guards, widens the rule's interval by g floors and returns meets or fails with the margin in floors, or a trial when a guard fires, no rule is qualified or the requirement lies inside the widened interval.

4 Demonstration on deep drawing

4.1 Alternatives and production data

The production data are the Real Deep Drawing and Cutting dataset (RDDAC) of the University of Stuttgart (Baum and Heinzelmann, 2026; Baum et al, 2026). Square blanks of DP600 steel, 210 mm wide and about 1 mm thick, were deep drawn into cups (OP10) and cut (OP20) on one tool set with two geometries, concave and convex side walls, which differ in base curvature, wall angle and bottom radius at once. Each geometry was run at three blank-holder forces (100, 300 and 500 kN) and three lubrication patterns (coarse, medium and fine, with series medians of 1.15–1.48, 0.92–1.23 and 0.85–1.13 g/m² of oil): 18 alternatives, each produced as one series of 500 consecutive parts, the change of geometry being the only change of product design. Every part has press signals, sheet and oil-film traverses and 3D scans after both operations; 10 of the 9,000 parts carry no scan. The series warm up by 1.0–5.0 K over their first 150 parts from punch temperatures of 20.5–29.6 °C, their median sheet thickness ranges from 0.986 to 0.993 mm, and the 100 kN series ran at 47 mm/s against 78–79 mm/s. The documentation states neither the order and dates of the series nor whether the parts were scanned in production order.

4.2 Simulations

The simulations come from the Deep Drawing and Cutting Simulations dataset (DDACS) of the same group (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025): LS-DYNA quarter models with nodes about 1 mm apart, a scaled DP600 flow curve, constant Coulomb friction and springback after both operations; the yield criterion and unloading model, which govern springback (Yoshida and Uemori, 2002), are not documented, and the stroke speed is not represented. For the RDDAC tools, 396 simulations vary the sheet thickness (0.95–1.00 mm) and the friction coefficient (0.05–0.15) at each geometry and force. A part is matched to the simulation with the nearest thickness and the friction coefficient that the dataset’s linear mapping assigns to its oil film, an assumption of the dataset’s authors on which Mc, M3 and every matched simulation depend. The nominal simulation of an alternative is at 0.99 mm, the grid value nearest the median thickness, and friction 0.10.

4.3 Quality characteristics and their measurement

Seven characteristics were extracted with the same definitions from the scans and the simulated nodes, referred to the plane of the cup bottom: the mid-side and corner draw-in and the flange waviness, process indicators on the flange that the cut removes, and four characteristics a drawing would carry: the wall angle (design angle 20° concave, 10° convex), the cup depth, and the arm angle and bottom dome of the cut part, the last two with requirements on absolute values. The four sides of a part measure several of them twice, which exposes the measurement: the flange width along the scan lines scatters 4 to 18 times more than across them, so the mid-side draw-in is taken across the scan lines, and the wall angle is the mean of the east and west walls (the north wall carries uncorrelated scatter, the south wall is shadowed). The mid-side draw-in

is quantised in steps of 0.019 mm (0.38 floors) by the pixel pitch. Differences between scans and simulations, such as the scanned surface against the shell mid-surface, enter the offsets between the two but not the floors.

4.4 Evaluation design

Every produced alternative is judged as a new design under grouped cross-validation (Roberts et al, 2017), one scheme per evidence relation. *New variant*: the alternative alone is held out and the other eight of its geometry calibrate, two siblings at its force among them. *New process setting*: the three alternatives of one force of a geometry are held out and the six others calibrate; 300 kN is interpolated, 100 and 500 kN extrapolated. *New family*: one geometry calibrates the rules for the other (two transfers). Pooled versions of the first two add the other geometry’s alternatives to the calibration set. Every calibrated rule is evaluated with and without the simulation (M1/M1n, M2/M2n, M3/M3n, M5/M5n), so that the paired difference isolates what the simulation contributes, and the nearest produced setting is the baseline without simulation and learning. A scope comprises 126 cases (18 alternatives, 7 characteristics).

The distances are computed exactly. Their intervals come from 1000 resamples of the held-out alternatives within each geometry, each with a block-bootstrap replicate of every true q_p , common to all rules so that differences get paired intervals; they hold the fitted rules fixed. A second bootstrap resamples whole lubrication patterns within each geometry, which keeps every case in its evidence relation, and refits every alternative-level rule (Table A2).

4.5 Computation and development record

The constants of the floor and the requirement grid were fixed before the decision results were seen, as the repository history records. The rules were not: M0–M5 were defined first; M0w, Mc, M1n, NN, M2n, M5n, the grouped schemes, the exact estimator and the noise floor of the Gaussian processes were added after the first results; and a review of that version added M1s, the jackknife+ intervals, the between-run noise floor and the analyses of Sections 5.4 and 5.7. The comparison of rules is therefore exploratory and carries the optimism of a selection made on the evaluation data (Cawley and Talbot, 2010); every variant tried is reported in the paper or the archived results. The pipeline reproduces every result from the cached feature tables in about four hours on four cores.

5 Results

5.1 What the production floor contains

In 95 % of batch pairs, two batches of 50 consecutive parts of one alternative differ by up to 0.051 mm in mid-side draw-in and 0.078° in wall angle (Table 3); one floor is 0.66 to 2.2 long-term standard deviations of the parts. The floors are set by drift within the run: without the time order they shrink by a factor of 2.1 to 4.5, batches one position apart differ by 0.51–0.80 floors and nine apart by 1.1–1.5, and the normal model of Section 3.2 reproduces the floors to within 12 %. Independent part-to-part noise,

Table 3 Production floor per characteristic, pooled and per geometry, with its components

Characteristic	F	95 % interval	Concave	Convex	F/σ_{LT}	σ_b/F	σ_w/F	Drift
Draw-in, mid-side (mm)	0.051	0.045–0.061	0.050	0.052	1.2/1.5	0.32/0.34	0.91/0.66	3.3
Draw-in, corner (mm)	0.046	0.035–0.052	0.053	0.030	1.7/1.3	0.31/0.35	0.65/0.77	4.5
Flange waviness (mm)	0.0028	0.0024–0.0031	0.0037	0.0021	2.1/0.8	0.33/0.30	0.55/1.31	2.5
Wall angle ($^\circ$)	0.078	0.066–0.10	0.081	0.078	1.6/0.7	0.32/0.26	0.68/1.39	2.1
Arm angle, cut ($^\circ$)	0.11	0.094–0.12	0.12	0.100	1.9/0.9	0.36/0.31	0.46/1.22	2.6
Cup depth (mm)	0.011	0.0096–0.012	0.0087	0.012	2.2/1.7	0.31/0.34	0.65/0.52	3.9
Bottom dome, cut (mm)	0.0070	0.0061–0.0081	0.0034	0.0082	1.4/1.5	0.33/0.36	0.76/0.55	3.9

Batches of 50 consecutive parts, 95 % quantile of the difference of batch centres; interval from 2000 resamples of batches within series. Ratios per geometry (concave/convex): σ_{LT} , median long-term standard deviation of the parts; σ_b , standard deviation between batch means corrected for $\sigma_w^2/50$; σ_w , within batches. Drift: floor over the floor of randomly permuted series.

scanner repeatability included, would produce 15–33 % of each floor, and the resolution is small against the floor except for the mid-side draw-in; assumption A3 holds for repeatability and resolution, but the stability of the scanner cannot be checked without a reference artefact. Nor do the process signals identify the source of the drift: batch centres of press force, punch temperature, sheet thickness and oil film predict it out of fold with R^2 of -0.23 to 0.20 , and removing the predicted drift changes the floors by -16 to $+12$ %. Variation between runs is not in the floor: the three lubrication series of one geometry and force differ by up to 1.2 (waviness) to 9.8 floors (dome), and a floor taken across them is 1.5 to 9.1 times the reference. The decided statistic reproduces to about one floor: the 95th percentiles of two batches of 100 parts differ by 0.92–1.49 floors in 95 % of pairs, so distances below about one floor are beyond what production, or a trial, can confirm.

5.2 What production resolves and how far the simulation misses it

All 63 geometry contrasts exceed the floor and blank-holder force is resolved in 105 of 126 contrasts, but lubrication only in 51 of 126 (42 with a bootstrap probability of at least 0.95), so a lubrication variant is often a near-replicate of its siblings. The minimum resolvable force change is 21–67 kN between 100 and 300 kN for most characteristics and lies beyond either force interval for the wall angle and the dome. The nominal simulation misses production by far more than a floor: its median offset reaches 109 floors, its force effect on the draw-in is 2.7 to 5.1 times the measured one, and the cut arms of the concave cups bend downwards in 89–100 % of the simulations but upwards in every part. Part of the offset is definitional: the cup depth differs by a median 0.90 and 0.95 mm, consistent with the scanned surface against the shell mid-surface, and removing this offset alone brings the decisive distance of the nominal simulation from 108 to 65 floors.

5.3 Decisions by evidence relation

Table 4 and Fig. 2 score the rules. The nominal simulation needs requirements 108 floors from the truth (bootstrap interval 101–115) before 95 % of its verdicts are

599 right; it errs mostly by rejecting adequate alternatives (safe distance 17 floors for false
600 accepts, 109 for false rejects).

601 *A new variant with a produced sibling.* With the other eight alternatives of the
602 family produced, every calibrated rule improves on the nominal simulation, and the
603 simplest is the most decisive: the nearest produced setting, the mean q_{95} of the two
604 siblings at the same force, is right beyond 3.4 floors (2.0–4.5), 0.17 mm of mid-side
605 draw-in. The scaled simulation follows at 4.9 floors; the Gaussian processes decide at
606 7.2 floors and are safe below one floor; the envelope rule is safe at 0.15 floors but
607 sends a third of the designs to trial at 10 floors and decides only at 23. Bias-correcting
608 the nominal simulation (M1, 15 floors) is worse than the median of the produced
609 alternatives without it (M1n, 9.1).

610 *A new process setting.* With every alternative at the new design's force held out,
611 all rules lose. The nearest produced setting is right beyond 8.5 floors (5.5–10.2); the
612 interval rules decide only at 13–31 floors, are safe at 5.9–16 and at 10 floors send 18–
613 37% of the designs to trial. Interpolating in force is much easier than extrapolating:
614 the nearest setting decides at 4.7 against 9.5 floors, and M2 and M5 are safe at 2.7
615 and 2.8 against 9.9 and 8.9 floors. The scaled simulation decides at 19 floors and,
616 extrapolating a fitted trend, falsely accepts failing alternatives as far as 164 floors from
617 their limit.

618 *A new design family.* In the two transfers every rule without the simulation fails
619 (131–133 floors), because nothing in one family's production record describes the other.
620 With the simulation, the bias correction decides at 35 floors, M5 at 38 and M3 at 49,
621 and the envelope rule is the safest at 16 floors; no rule is safe closer than that. Scored
622 in the floor of the produced family, which a design team would have, M5 decides at
623 35 and is safe at 31 floors and the envelope rule at 41 and 26. Sixteen floors are about
624 0.8 mm of mid-side draw-in or 1.3° of wall angle.

625

626 5.4 Where the distances come from

627

628 Three features of the evaluation design shape the distances. First, the sibling relation
629 mixes replication with variation: in 58 of the 126 new-variant cases neither sibling
630 differs resolvably from the held-out alternative, and the nearest produced setting is
631 then right beyond 0.85 floors, the reproducibility of q_{95} between near-replicate runs;
632 when both differ resolvably (34 cases), it needs 4.9 floors and M5 9.7. Second, the
633 extrapolation cases are confounded with stroke speed: the production-only rules lose
634 almost only at 100 kN, where the press also ran slower (the nearest setting decides at
635 11 floors against 4.9 at 500 kN), whereas the simulation-based interval rules are unsafe
636 at both ends. Third, the production record of the other family does not help within a
637 family: pooling it makes most rules less decisive (M5 from 7.2 to 11 floors for a new
638 variant, M2 from 23 to 53, M2n from 19 to 325), although it raises the coverage of M5
639 from 86 to 90% and makes the centre of its interval the most accurate point prediction
640 within the family (2.9 floors against 3.4 for the nearest setting).

641 The relation therefore does not govern decision fitness in general. Over the eleven
642 calibrated rules and three relations, the relation explains 69% of the variation of the
643 log decisive distance and the rule 6%, but this share comes from the new family, where
644 the rules without the simulation fail by construction; within a family the rule explains

Table 4 Decisive and safe distances (floors) by the relation of the new design to the produced evidence

Rule	New variant		New process setting				New family		Pooled	
	δ_d/δ_s	FA	δ_d/δ_s	FA	interpolated	extrapolated	δ_d/δ_s	FA	variant	setting
M0	108	17	108	17	109	106	108	17	108	108
M1	15	15	24	24	11	25	35	33	27	28
M1s	4.9	4.7	19	164	19	19	122	122	15	20
interval	4.7–6.3		14–21		19–20	15–19	121–123			
M2	23/0.15	0.30	18/8.1	8.3	18/2.7	18/9.9	51/16	11	53/0	53/1.0
M5	7.2/0.75	0.15	19/6.8	7.6	19/2.8	19/8.9	38/28	32	11/0.05	28/5.4
interval	6.6–8.4		18–20		18–20	18–20	35–39			
M5n	7.2/0.55	0.10	17/6.1	9.0	16/0	18/8.6	132/130	167	9.8/0.20	16/2.7
interval	6.4–7.5		16–18		15–16	16–19	130–133			
NN	3.4	2.8	8.5	10	4.7	9.5	131	169	3.4	8.5
interval	2.0–4.5		5.5–10		4.3–5.1	6.6–11	129–131			
M0w	119/94	9.2	119/94	9.2	119/94	119/77	119/94	9.2	119/94	119/94
Mc	27	28	39	39	14	40	38	42	32	33
M3	23/0	0	31/16	13	29/13	33/17	49/21	20	25/0	32/5.1
M1n	9.1	10	10	12	4.8	12	131	173	130	130
M2n	19/0	0.20	15/5.9	9.2	13/0	16/9.1	132/129	165	325/0	325/4.0
M3n	11/0.05	0	13/7.4	9.2	15/7.0	13/7.8	133/128	164	15/0	15/4.9
M6	17/0.05	0	56/6.2	5.8	47/0	56/8.1	41/25	29	–	–

Over all seven characteristics; one number for rules that always decide. FA: safe distance on the false-accept side (failing alternatives). Interval: bootstrap 95 % interval of the decisive distance (fits fixed). 126 cases per relation (42 interpolated, 84 extrapolated); a new family rests on two transfers. Pooled: the other geometry’s alternatives added to the calibration set. Below the line, the further rules; M6, the M5 mean with a conformal margin on standardised leave-one-out residuals, appears only in the robustness analysis.

68 % and the relation 15 %. The best attainable distance still grows with the novelty of the design, from 3.4 floors with a sibling to 8.5 for a new setting and 35 for a new family.

5.5 What the simulation and learning contribute

Pairing each calibrated rule with its counterpart without the simulation isolates what the simulation adds as it is used. Added as an offset, it adds nothing within a family and degrades two rules: M5 decides where M5n does (difference 0.0 floors, -0.4 to $+1.1$), and M2 and M1 decide 3.9 ($+0.5$ to $+7.4$) and 6.2 floors ($+4.6$ to $+9.7$) further out than M2n and M1n, because the simulated force effect is too large. Rescaled, it adds information within a family, where the scaled simulation decides 4.2 floors closer than the median of the produced alternatives (-5.4 to -0.5), though 1.5 floors less close than the nearest setting; for a new setting the rescaled trend is worse than none. Only for a new family is the simulation decisive, by about 94 floors, and even there production knowledge carries over where a drawing nominal transfers: for the wall angle, the design angle plus the produced family’s deviation from its own design angle decides at 2.0 floors, against 7.3 for M5. A surrogate of the simulations reproduces them to within 1.8 floors for 12 of the 14 characteristics and geometries, yet its decision fitness is that of the simulation it reproduces.

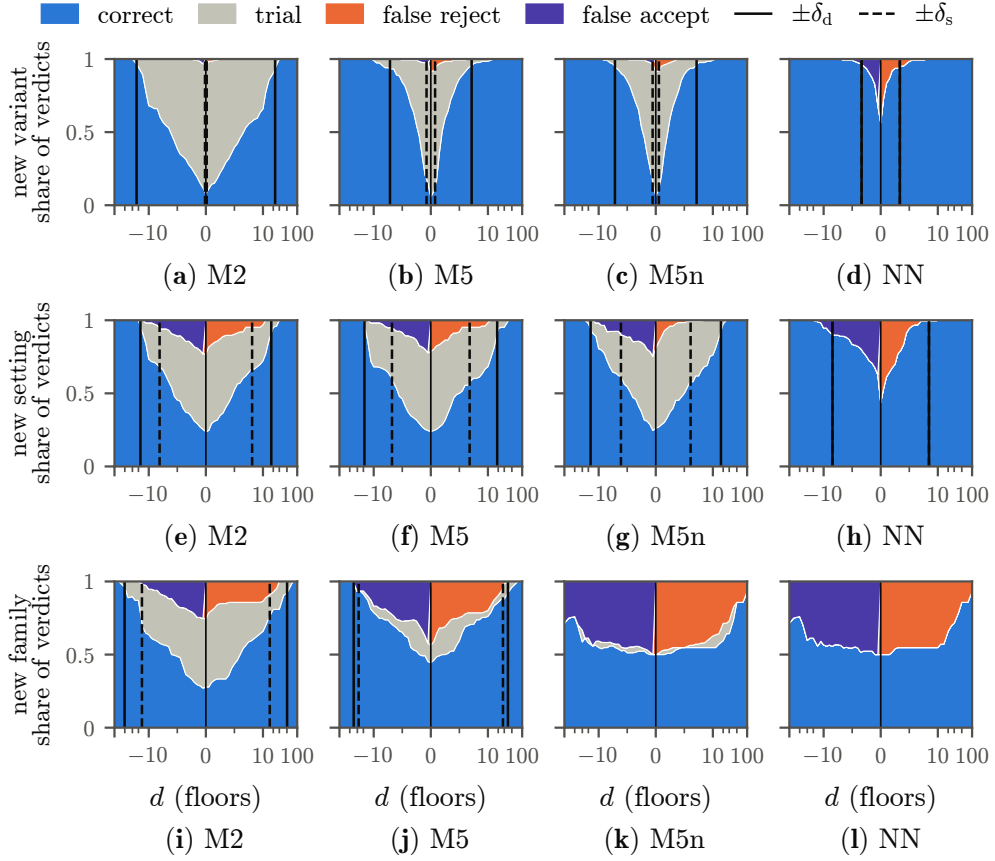


Fig. 2 Verdict shares against the requirement distance d (symmetric logarithmic axis, linear within ± 10 floors; 126 cases per panel) with the decisive and safe distances marked

Learning adds intervals rather than accuracy. Used as point rules, the centres of the M5, M5n and M3n intervals decide at 3.3, 3.1 and 3.7 floors within the family, as close as the nearest setting, so the ranking of the learned rules rests on the width of their intervals. With the noise floor set by the variation between runs, the Gaussian-process intervals cover 85–86 % of the true q_{95} within the family against a nominal 90 %, the largest-residual margin 89 % (it guarantees at least $(n-1)/(n+1) = 7/9$ under exchangeability, which these cases lack) and the jackknife+ interval of M3 96 %; for a new setting the calibrated intervals cover 43–55 %, and for a new family the Gaussian processes at most 12 %. The inductive biases explain much of this: trees do not extrapolate, two force levels identify at most a linear trend, and in the median new-setting fit the length scale of the force sits at its lower bound, so the Gaussian process reverts to its mean.

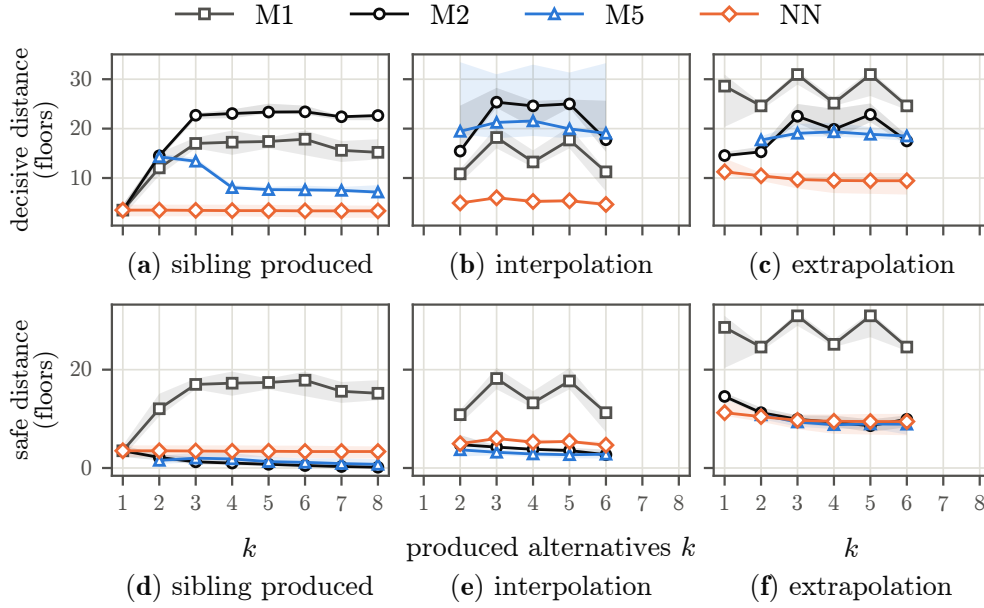


Fig. 3 Decisive and safe distances against the number of produced alternatives of the family, by the relation of the new design to them; bands: 95% intervals over the held-out alternatives

5.6 Calibration budget

Figure 3 shows how the distances depend on the number of produced alternatives of the family and on their relation to the new design. A single produced sibling decides: with one alternative at the new design's force, the bias correction, the envelope rule and the nearest setting coincide by construction (the simulation terms cancel and a single residual gives no margin) and decide at 3.5 floors (2.3–4.6) without a wrong verdict at 10 floors. Further alternatives do not improve the nearest setting; they make the bias correction worse (12–18 floors), the envelope rule safer but less decisive, and the Gaussian processes safer and more decisive (from 12–14 floors with two alternatives to 7.2 with eight). Without a sibling, bracketing the new force buys safety but not decisiveness: from two alternatives on, the interval rules are safe at 0–4.8 floors with no wrong verdict at 10 floors but decide only at 15–25 floors, against 4.7–6.0 for the nearest setting. Extrapolation stays unsafe at every budget. Averaged over all subsets, the distances shrink with the budget mainly because the share of subsets containing a sibling grows, from 46% with two alternatives to 96% with six; the odd–even pattern of the bias correction comes from its median offset, which averages two forces only for an even number of alternatives. With calibration series of 50, 100 or 250 parts instead of 500, no distance within a family moves by more than 1.2 floors, and for a new family by at most 4 floors (Table A2): the qualification needs short runs of several variants, and only the floor needs a run long enough to show the drift.

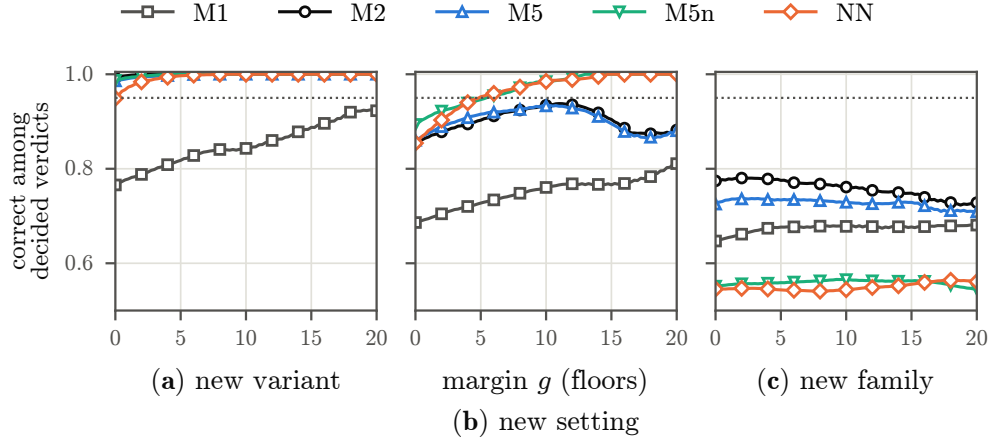


Fig. 4 Share of correct verdicts among the decided ones against the observable margin between interval and requirement, for requirements within 20 floors of the truth

5.7 The procedure's decision rule

What a designer observes for a single verdict is the margin between the predicted interval and the requirement, and its value as evidence depends on the relation (Fig. 4): for a new variant at least 95 % of the decided verdicts of the interval rules are correct at every margin; for a new setting the share grows with the margin, for the nearest setting from 85 % to 95 % beyond 5 floors; for a new family M5 is right in 71–74 % of its decided verdicts whatever the margin. Step 5 turns this into guard bands: none for the interval rules within the family and 0.25 floors for the nearest setting; 2.0 floors for an interpolated setting, with which the nearest setting decides at 6.7 floors, is safe at 2.7 and sends no design to trial at 10 floors; 7.0 floors for an extrapolated one, with which it decides at 17 floors and sends a quarter of the designs to trial at 10 floors; and for a new family no guard band up to 20 floors reaches 95 %, so every design goes to trial. The guard bands are chosen on the evaluation data, which makes the trial rates optimistic. For a new setting the range guard refuses exactly the extrapolated forces, after which M2, M5 and the nearest setting make no wrong verdict at 10 floors; the novelty guard also refuses many cases that would have been decided correctly.

A worked example. A tryout engineer has to decide whether the concave cup at 300 kN with medium lubrication, a new process setting, can be released without a trial run, with the six alternatives at 100 and 500 kN produced and a requirement that 95 % of the parts have a mid-side draw-in of at most 15.30 mm. The floor is 0.050 mm, the relation is interpolation and no guard fires. The nominal simulation predicts 15.66 mm and fails the design. The nearest produced setting predicts 14.99 mm, a margin of 6.2 floors, more than its guard band of 2.0 floors, so step 6 returns meets; beyond a margin of 6 floors every decided verdict of this rule for an interpolated setting was correct. M2, M5 and M5n return intervals reaching 15.37, 15.58 and 15.33 mm and send the design to trial. The 500 parts had a 95th percentile of 14.88 mm, 8.4 floors below the limit: the procedure was right, the simulation wrong and the interval rules cautious.

5.8 Requirements anchored in capability and in tolerances

Upper limits at which an alternative has a performance index of 1.0, 1.33, 1.67 or 2.0 lie a median 1.0, 1.7, 2.5 and 3.1 floors above its 95th percentile, inside every decisive distance; every alternative meets them, so a verdict is right, a false reject or a trial. At an index of 1.33 the nearest setting still says meets in 91 % of the new-variant cases, because most of its errors are far smaller than its 95 % distance, whereas M5 says so in 44 % and sends the rest to trial; for a new setting the two are right in 65 and 32 %, and for a new family no rule exceeds 54 %. Release-level requirements are therefore decided at the design stage only within a family, and then by the production record. General angular tolerances of ISO 2768-1 (ISO, 1989) give a coarser scenario that mostly tests whether a rule recognises gross failures (32 of its 108 decisions lie within 5 floors of the limit). With a false reject at half the cost of a false accept, its cheapest rule is the nearest setting at any cost for a new variant; for a new setting the production-only interval rules while a trial costs at most a tenth of a false accept, and the nearest setting beyond; for a new family a trial of every design while a trial costs at most a tenth, the envelope rule at a fifth and M5 beyond.

5.9 Robustness

The analysis was repeated under variants of the data, the constants and the model specifications (Table A2). Adjusting for punch temperature changes no calibrated distance by more than 2.6 floors. Dropping the warm-up of the first 150 parts shrinks the floors and so lengthens the distances (the nearest setting 4.1 floors with a sibling, 12 for a new setting), and floors taken between series, 1.5 to 9.1 times the reference, shorten them (0.75 and 2.7 floors). Other floor protocols rescale the distances by 0.60 to 1.42, the Gaussian-process specifications move them by at most 2.1 floors, and in the unit of the reproducibility of q_{95} the nearest setting decides at 2.8 and 6.9 floors. In every variant that includes it, the nearest setting remains the most decisive rule within a family and the envelope rule the safest for a new family. The refitting bootstrap widens the intervals of the learned rules most (M5 4.0–11 floors for a new variant, the nearest setting 2.3–4.6), yet the nearest setting is the more decisive in all 200 resamples, against M5 and M5n and for a new setting alike.

6 Discussion

6.1 What the evaluation establishes

Five findings answer the research questions. Two follow from the definitions and hold wherever the framework is applied: the floor puts every rule and characteristic on a common production-referenced scale, though not on a common scale of nonconforming fractions, and as a within-run quantity it is a lower bound of the variation of series production (RQ1); and an interval rule trades decisiveness for safety, which the two distances make explicit, and reaches its nominal level only where the calibration alternatives are exchangeable with the new design. Three are observations on this dataset and hypotheses for confirmation: the best attainable distance grows with the novelty

Table 5 Guidance for design-stage manufacturability decisions, with its basis

Situation	Guidance	Basis
<i>From the framework (general)</i>		
Before trusting a predictor	Measure the floor on consecutive production of the family with a stated protocol and report σ_b , σ_w and its lag dependence; treat it as a lower bound of series variation.	definitions
Qualifying a rule	Evaluate it for the evidence relation of the coming designs by grouped cross-validation, and split near-replicate siblings from resolvable ones.	definitions
Choosing a rule	Compare it with the nearest produced setting and with its own version without the simulation, on decisive and safe distance jointly or by expected cost.	definitions
Deciding a single design	Apply the rule's guard band to the observable margin; a margin inside the guard band, or a fired guard, means a trial.	definitions
Fixing a requirement	If the expected margin to the capability is below the safe distance of the best qualified rule, plan a trial; no decision is reliable closer than a trial reproduces the quantile.	definitions
<i>Observed on this dataset (18 alternatives, 2 geometries, 7 characteristics)</i>		
New variant with a produced sibling	nearest produced setting right beyond 3.4 floors (0.85 for a near-replicate); calibrated interval rules safe below one floor	126 cases
New process setting	nearest produced setting right beyond 4.7 floors interpolated and 9.5 extrapolated; interval rules safe at 5.9–16 floors	42 + 84 cases
New family	best rule safe at 16 floors; rules without the simulation fail; a transferable drawing nominal helps	two transfers
Release-level requirements	lie within 1–3 floors of the 95th percentile, inside every decisive distance	126 cases

of the design, while within a family the choice of rule matters more than the relation (RQ2); production evidence buys safety before decisiveness, since interval rules become safe with a few produced alternatives that bracket the new design and decisive only with near-replicates; and with this simulator and six to nine produced settings per fit, the simulation contributed as an offset only across families, a fitted scale made it useful within a family but harmful for a new setting, and learned models predicted as well as the nearest produced setting while adding abstention (RQ3).

6.2 Using the framework in design

Table 5 separates guidance that follows from the framework from observations on this dataset. Before a requirement is fixed, the expected margin between requirement and capability can be compared with the safe distance of the best rule qualified for the relation of the design; if the margin is smaller, a trial or further production evidence has to be planned. Release-level requirements make this the usual case: they lie within a few floors of the capability, where only a near-replicate production record decides reliably, consistent with the practice of requiring a production trial run for a process change. For a single design, step 6 gives a verdict whose guard band a designer can see, and the worked example shows that the most decisive rule is also the most transparent: the nearest produced setting points to the produced alternative it copies, whereas a Gaussian-process interval does not. Which rule to prefer depends on the costs, which

differ by relation: a trial of a new force on an existing tool is a short press run, a trial of a new family may need a new tool. In a company, steps 1 to 5 fit the start of production or the production part approval of a family, when its series are measured anyway, and step 6 the later decisions on variants and process changes.

6.3 Simulation and machine learning in this regime

Within a design family the production record carried the decision. The simulated force trend was 2.7 to 5.1 times too large, and calibrating the friction coefficient did not repair it, which points to the friction law and the blank-holder model rather than to a parameter. Learned models added an interval that is safe where the calibration alternatives are exchangeable with the new design and over-confident where they are not. A multi-task model cannot help a family with no produced alternative, because the correlation between families cannot be estimated without its data; the pooled Gaussian process, which carries the geometry as an input, is the borrowing of strength the data allow, and it improved coverage but not decisiveness. These findings concern one simulator of low fidelity for this production, used through offsets and one scale, and learners with six to nine produced settings each; a surrogate of these simulations inherits the simulator's distance, whatever its accuracy against the simulation. For the evaluation of AI-based DFM support the lesson is methodological: held-out accuracy is not enough; a learned model should be compared with a production-record baseline, scored by evidence relation in a unit that production defines, and shown to its user with a guard band. Whether that makes designers' reliance match a rule's reliability (Lee and See, 2004), and when an abstention should go to an expert rather than to a trial (Madras et al, 2018), are hypotheses for a study with design teams.

6.4 Implications for design research

The framework gives several constructs of design research a production-referenced measure. For variant, adaptive and platform design (Pahl et al, 2007; Simpson et al, 2006), the distances by evidence relation quantify how far production knowledge carries from produced variants to a new one, with the nearest produced setting as the baseline that any capability database (Thornton, 2004) or learned model has to beat. The safe distance is the part of a margin that covers predictive uncertainty, as distinct from the part that absorbs change (Eckert and Isaksson, 2017; Eckert et al, 2019). For testing strategies (Thomke and Bell, 2001; Salado and Kannan, 2018; Camburn et al, 2017), overlapped development (Krishnan et al, 1997; Terwiesch et al, 2002) and set-based design (Sobek et al, 1999), the distances say when a test is cheaper than a decision, when design-stage information is fit to act on and how wide a feasibility limit has to be. For validation, the evidence relation is the question of the application domain (Oberkampff and Roy, 2010), and the distances could serve as evidence of a risk-informed credibility assessment (ASME, 2018). In the terms of the design research methodology (Blessing and Chakrabarti, 2009), the evaluation is a support evaluation: it shows structural validity in the sense of the validation square, that the metrics discriminate between rules and relations on real data (Seepersad et al, 2006), and is a first step towards performance validity, which would require showing that decisions taken with the framework are better than decisions taken without it.

Table 6 What the framework requires, mapped onto other processes

Process	Batch and run	Dominant variation	Evidence relations	Floor
Sheet forming (this paper)	consecutive parts of one series	drift within a run; coil, die state	variant, setting, family	batch centres within a run
Injection moulding	shots of one cavity and set-up	cavity, shot, set-up	cavity, process window, new mould	within cavity, between cavities
Machining	consecutive parts between offsets	tool wear, compensated by offsets	tool path, cutting data, new part	between offset intervals
Casting	castings of one pattern and melt	melt, pattern position	alloy, gating, new pattern	between melts
Additive manufacturing	replicated specimens of one build	build, position, orientation	orientation, parameters, new part	between builds, via anchors

6.5 Reuse for other processes and AI-based DFM tools

The framework needs several produced variants of a design family whose relations to each other span those of the coming designs, a series of consecutive parts or replicated batches of one design per variant, a design-stage predictor for every variant, and characteristics measured identically on parts and predictions. Here the long series served the floor, whereas calibration runs of about 50 parts per variant qualified the rules as well as full series; how many long runs a floor needs was not tested. Table 6 maps these requirements onto the processes named most often in DFM; attribute outcomes such as splits, porosity or short shots need a requirement on a rate rather than a quantile, which the framework does not yet cover. A predictor need not return an interval: any tool that returns meets, fails or trial for a stated requirement, such as a classifier at its operating threshold or a language-model check given the requirement value, can be scored by sweeping the requirement, and for continuous design spaces the relation becomes a distance from the produced designs in descriptor space or the position relative to their hull. A minimal report comprises the floor with its protocol and variance components, the decisive and safe distances by relation with refitting intervals and one-sided values, the coverage of interval rules, the nearest-produced-setting and no-simulation baselines, and the guard bands and guard performance. Distances per characteristic rest on as many cases as there are produced alternatives, so a team would pool characteristics with similar ratios of floor to part scatter, or accumulate evaluations over families. An attempt to apply the framework to an open dataset of powder bed fusion of PA12 (Leirimo, 2021) failed on the second requirement: grouping specimens by orientation class mixed orientations in a batch, and eight identical anchor specimens per build were too few for a floor; a between-build floor from replicated anchors would be the remedy.

6.6 Threats to validity

The evaluation covers one material, one press and two geometries, and the relations within a family come from a 3×3 factorial in which lubrication is mostly not resolved; the procedure is general, the numbers are not. The new family rests on two transfers between geometries that differ in three features at once, and the intervals of its

distances hold the two fits fixed. Each alternative was produced once, so the floor covers drift within a run but not between days, coils or tool states, the reference q_{95} is that of one run, and the order and dates of the series are unknown; between-series variation bounds the missing component at 1.2–9.8 floors (Section 5.1). Scanner drift cannot be separated from process drift. The extrapolation in force is confounded with stroke speed, which the simulation does not represent, so the extrapolation results describe a force and speed change together. The requirements are one-sided limits on the 95th percentile of a characteristic averaged over sides, below the conformance of series release. Several rules, the grouped schemes and the guard bands were added or chosen after earlier results had been seen and are evaluated on the same data, so the comparison between rules carries selection optimism (Cawley and Talbot, 2010); the analysis was carried out by a single analyst, with code and data open for replication.

7 Conclusions

Design-stage manufacturability decisions can be evaluated against production on a scale that production defines: the decisive and safe distances in floors tell how close to a requirement a rule decides and how close it can be relied on not to mislead, and a guard band turns them into a rule for a single decision.

On 18 deep-drawn design–process alternatives with matched simulations, the best attainable distance grew with the novelty of the design: 3.4 floors with a produced sibling, 8.5 for a new process setting, and no rule safe closer than 16 floors for a new family. Within a family the choice of rule mattered more than the relation, release-level requirements lay inside every decisive distance, and in this small-data regime the simulation as used, a model with a large and partly sign-inconsistent bias, contributed only across families, while learned models added abstention rather than accuracy.

Four extensions follow: series produced on several days, coils and tool states, so that the floor covers variation between runs; design families with continuous geometric descriptors; a confirmatory evaluation of the frozen procedure on another dataset or process; and a study with design teams of whether decisions taken with the framework are better. The released code, which reproduces every result from the public data, is the starting point.

Appendix A Notation, robustness and computation details

Table A1 summarises the notation of Section 3, and Table A2 gives the distances under the robustness variants; the other results quoted in the text, such as guard bands, coverage, paired differences and distances per characteristic, are written by the released code and archived with it.

Computation details. Batches need at least 80 % valid values. M5 and M5n are Gaussian-process regressions (constant times anisotropic squared-exponential kernel plus white noise, normalised targets, three restarts, length scales in $[0.1, 100]$) on a geometry indicator, the force in 100 kN and the lubrication rank. NN takes the mean q_{95} at the nearest force of the family, both neighbours when equally near. Mc calibrates,

1059 **Table A1** Notation and terms

1060	Symbol, term	Meaning
1061		
1062	$a, c, q_p(a, c)$	alternative (design–process combination), characteristic, p -quantile of the parts of a ($p = 0.95$)
1063	R, d	requirement (upper limit on c or $ c $); its distance from the truth in floors, $R = q_p + dF_c$
1064	$[L_r, U_r], V_r$	interval of rule r for q_p ; verdict meets, fails or trial (Eq. 1)
1065	B, m, F_c	batch size (50); batch centre (10% trimmed mean); production floor (Eq. 2)
1066	σ_b, σ_w	standard deviation between batch means and between parts within a batch
1067	ESR, MRC	effect-to-scatter ratio (Eq. 3); minimum resolvable change, $F_c/ s_c $
1068	u, v	distances in floors by which an interval excludes the truth from above and below
1069	$\kappa(\delta), \omega(\delta)$	shares of correct and wrong verdicts among the feasible verdicts at $ d = \delta$
1070	δ_d, δ_s	decisive distance ($\kappa \geq 0.95$ beyond it) and safe distance ($\omega \leq 0.05$ beyond it)
1071	FA, FR side	one-sided distances for failing alternatives (false accepts) and adequate ones (false rejects)
1072	g	guard band: observable margin beyond which at least 95% of the decided verdicts were correct
1073	Calibration alternatives	produced alternatives whose record calibrates a rule
1074	Evidence relation	sibling (same setting produced), new process setting (interpolated, extrapolated), new family
1075	Calibration budget	distances against the number k of produced alternatives, by relation
1076	Guard	family, range and novelty checks that send a design outside the calibration to trial
1077		
1078		
1079		
1080		

1081 per lubrication pattern, the friction coefficient whose simulated draw-ins are closest
1082 to the calibration centres. M3 and M3n use histogram gradient boosting (depth 3,
1083 200 iterations, rate 0.08) on geometry, force, lubrication, sheet thickness and oil film,
1084 with 500 incoming conditions drawn from calibration parts of the same pattern; their
1085 jackknife+ interval combines the leave-one-out predictions for the new alternative with
1086 the absolute leave-one-out residuals about their median. Guard bands lie on a grid
1087 of 0.25 floors. The refitting bootstrap draws the three lubrication patterns of each
1088 geometry with replacement 200 times, at least two of them distinct. Software: Python
1089 3.11 with NumPy, pandas, SciPy, scikit-learn (Pedregosa et al, 2011) and Matplotlib.

1091 **Acknowledgements.** The author thanks the authors of the RDDAC and DDACS
1092 datasets at the University of Stuttgart for publishing the data on which this work rests.

1093

1094 Declarations

1095

1096 **Funding.** No specific funding was received for this work.

1097 **Competing interests.** The author declares no competing interests.

1098

1099 **Ethics approval and consent.** Not applicable.

1100 **Data availability.** All data are public. RDDAC: doi:10.18419/DARUS-5589 (ver-
1101 sion 2.0); DDACS: doi:10.18419/DARUS-4801 (version 3.0); both licensed CC BY 4.0.
1102 The extracted per-part and per-simulation feature tables are provided with the code.

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Table A2 Decisive/safe distances (floors) under the robustness variants

Variant	New variant				New setting			New family	
	M1s	M2	M5	NN	M2	M5	NN	M2	M5
Reference	4.9	23/0.15	7.2/0.75	3.4	18/8.1	19/6.8	8.5	51/16	38/28
Temperature-adjusted	5.0	23/0.10	7.2/0.65	3.4	17/7.6	19/6.8	9.4	48/15	37/27
First 150 parts dropped	6.5	27/0.20	9.0/0.45	4.1	21/8.7	24/8.7	12	63/18	39/29
Floor between series	2.9	11/0	1.3/0.20	0.75	9.9/2.6	11/2.3	2.7	16/3.8	12/9.3
Floor: batches of 25	4.6	21/0.15	6.4/0.70	2.9	16/6.9	17/6.3	7.7	43/14	31/23
Floor: batches of 100	5.3	24/0.15	8.2/0.80	3.5	19/8.5	21/7.9	8.9	57/17	46/29
Floor: 90 % quantile	6.2	31/0.20	8.8/0.90	4.0	21/10	23/8.6	10	61/20	45/34
Floor: 99 % quantile	3.4	14/0.10	5.2/0.50	2.4	11/4.4	13/4.4	5.7	37/11	29/15
Calibration series of 50	5.0	22/0.20	6.8/0.70	3.4	18/7.5	19/6.2	8.4	50/16	37/28
Calibration series of 100	4.9	22/0.40	6.8/0.40	3.3	18/7.5	19/6.2	8.7	50/16	38/28
Calibration series of 250	4.9	22/0.45	7.0/0.45	3.3	18/7.5	19/6.2	8.7	50/16	38/28
Conformance $p = 0.90$	4.9	23/0.20	7.2/0.70	3.3	18/7.5	18/6.4	8.4	51/15	38/28
Conformance $p = 0.99$	5.1	23/0	7.5/0.70	3.3	18/8.3	20/6.9	8.7	51/16	38/28
Interval level 0.80		23/0.15	6.4/1.0		18/8.1	17/8.1		50/17	38/28
Interval level 0.95		23/0.15	8.0/0.25		18/8.1	20/6.2		51/16	39/27
Response surface	4.8	25/0.30	7.2/0.70	3.4	18/7.6	20/6.9	8.5	49/25	37/27
GP: sampling noise floor			6.7/1.4			18/7.4			38/30
GP: no noise floor			6.7/1.5			18/7.4			38/30
Unit: q_{95} reproducibility	5.3	17/0.15	6.9/0.40	2.8	15/6.0	18/6.2	6.9	41/13	34/30
Refit bootstrap, δ_d	4.9–7.3	22–23	4.0–11	2.3–4.6	16–18	18–20	6.9–9.2	49–53	34–39

Rules refitted for every variant; empty cells: the variant does not affect the rule. Floor variants and the unit of q_{95} reproducibility (95 % quantile of the difference between the q_{95} of two batches of 100 parts) rescale the distances. Calibration series of n : the calibration alternatives' q_{95} from their first n parts. Conformance p : the decided statistic is the p -quantile. Response surface: nominal simulation and envelope from a quadratic fit over thickness and friction. Other kernels and descriptors of the GPs move M5 by at most 0.4 floors. Refit bootstrap: 95 % interval when the lubrication patterns of each geometry are resampled and every rule is refitted (200 resamples).

Code availability. The complete analysis pipeline is released under an open licence at <https://github.com/htcanseven/Claude> (directory `scripts/`) and reproduces every number in this paper from the public data; a versioned archive of the code and the feature tables is deposited on Zenodo [DOI pending].

Author contributions. H.T.C. conceived the method, designed and carried out the analysis, and wrote the paper.

Use of generative AI. A large language model (Claude, Anthropic) was used under the author's direction for code development, literature retrieval, data analysis, drafting and editing, and simulated peer review of earlier versions. The author designed the study, verified all analyses, checked every reference against its published record, and takes full responsibility for the content. All figures are produced by the released code from the data; no AI-generated images are used.

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